Academic year 2025-2026 (Odd Sem)

DEPARTMENT OF
COMPUTER SCIENCE AND ENGINEERING

Date	13th December 2025	**Maximum Marks**	10+50
Course Code	CD2521A	**Duration**	120 Minutes
Sem	V	Faculty: PD /PT/AKB/NGD/SGD/CNS/SNM/VMN/KSK/Harshitha V	

Database Management Systems (Common to CS, IS, CD, AI & CY)

Sl.No	PART-A	M	BT	CO
1.	Differentiate between SELECT and PROJECT operation with an example .	2	L3	2
2.	State Boy codd Normal form .	2	L2	2
3.	Mention the three types of JOIN operation in Relational Algebra .	2	L3	2
4.	Write reflexive, decomposition/projective inference rules for functional dependencies .	2	L2	2
5.	What is the difference between DROP and TRUNCATE .Write one command to add a new column 'email' to the Employee table.	2	L2	1
Sl. No.	**PART-B**	**M**	**BT**	**CO**

1		Consider the following database schema and write the SQL queries for each of the tasks listed below: Student (student_id, sname, major, GPA) Faculty (facult_id, fname, dept., designation, scr) Course (course_id, cname, faculty_id) Enroll (course_id, student_id, grade) Write SQL queries for the following: 1. List the names of all students enrolled for the course 'CS_53'. 2. List the names of students enrolled for 'CS_53' and who have received an 'A' grade. 3. List all departments having an average salary above Rs. 20,000. 4. List the names of faculty members beginning with 'R' and ending with 'U'. 5) List courses along with the number of students enrolled, but display only those having more than 25 students	10	L3	2
2		Consider the relational database of the following schema. *LIVES*(*p_name, street, city*) *WORKS*(*p_name, c_name, salary*) *LOCATION* (*c_name, city*) *MANAGES* (*p_name, mgr_name*) Give an expression in the relational algebra for each query: i) Retrieve all person names who live in *BANGALORE*. ii) Retrieve the name and salary of all the persons who work for the company *'HP'*. iii) List the names of the persons working for *HP* along with the cities they live in. iv) Find the names of the persons who live and work in the same city. v) Retrieve the city of *INFOSYS* company. vi) Display all the employees who live in *"Bangalore"* and works for "*INFOSYS*".	10	L3	2

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3	(a)	Explain 2NF and 3NF with examples.	06	L2	1
	(b)	Explain DIVISION operation of relational algebra with an example .	04	L2	1

4	(a)	Consider the relation scheme $R = (A,B,C,D,E,F,G,H,I,J,K,L)$ with the set of functional dependencies $F = \{\{A,B\} \rightarrow \{C\}, \{B\} \rightarrow \{F\}, \{F\} \rightarrow \{G,H\}, \{D\} \rightarrow \{I,J\}, \{B\} \rightarrow \{K,L\}\}$ and keys $\{ABD\}, \{K\}$. Find the $FD's$ which does not satisfy $2NF$ test. Decompose R into 2NF relations.	6	L4	2
	(b)	Explain INSERT ,DELETE and UPDATE anomalies with example	4		
5	(a)	Consider the following schema and obtain the SQL queries. *Employee* (*fname, lname, SSN, bdate, address, sal, SUPERSSN, dno*) *Project* (*fpname, Pnumber, Ploaction, dnum*) *Workson* (*essn, pno, hours*) *Department* (*dname, dnumber, mgrssn, mgsstartdate*) i) Find the sum of salaries of all employees, the maximum salary, minimum salary. ii) Retrieve the total number of employees in the company and number of employees in Research dept. iii) For each dept., retrieve dept. number, number of employees in dept. and their average salary.	6	L4	2
	(b)	Consider the following functional dependencies $SSN \rightarrow Ename$, Pnumber → {Pname, Plocation} $\{SSN, Pnumber\} \rightarrow hours$ Find the closure for the set of attributes under this set of FD's. $\{SSN\}+ = ?$ $\{SSN, Pnumber\}+ = ?$ $Pnumber+ = ?$	4	L3	

COURSE OUTCOMES:

CO1: Understand and explore the needs and concepts of relational, NoSQL database and Distributed Architecture **CO2:** Apply the knowledge of logical database design principles to real time issues.

CO3: Analyse and design data base systems using relational, NoSQL and Big Data concepts

CO4: Develop applications using relational and NoSQL database

CO5: Demonstrate database applications using various technologies.

COs/BTL	CO1	CO2	CO3	CO4	CO5	L1	L2	L3	L4
Marks	12	38	--	--	--	--	20	20	20